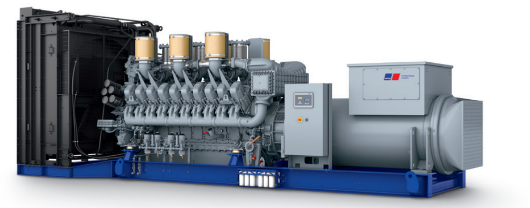
Diesel Generator Set

mtu 20V4000 DS3300

380V – 11 kV/50 Hz/grid stability power/ fuel consumption optimized/20V4000G34F/water charge air cooling



Optional equipment and finishing shown. Standard may vary.

Product highlights

Benefits
— Low fuel consumption
— Optimized system integration ability
— High reliability
— High availability of power
— Long maintenance intervals

Support
— Global product support offered

Standards
— Engine-generator set is designed and manufactured in facilities certified to standards ISO 2008:9001 and ISO 2004:14001
— Generator set complies to ISO 8528
— Generator meets NEMA MG1, BS 5000, ISO, DIN EN and IEC standards
— NFPA 110

Power rating
— System ratings: 3020 kVA - 3130 kVA
— Accepts rated load in one step per NFPA 110*
— Generator set complies to G3 according to ISO 8528-5
— Generator set exceeds load steps according to ISO 8528-5*

Performance assurance certification (PAC)
— Engine-generator set tested to ISO 8528-5 for transient response
— 100% load factor
— Verified product design, quality and performance integrity
— All engine systems are prototype and factory tested

Complete range of accessories available
— Control panel
— Power panel
— Circuit breaker/power distribution
— Fuel system
— Fuel connections with shut-off valve mounted to base frame
— Starting/charging system
— Exhaust system
— Mechanical and electrical driven radiators
— Medium and oversized voltage alternators

Emissions
— Fuel consumption optimized

Certifications
— CE certification option
— Unit certificate acc. to VDE-AR-N 4110

* Changes to the standard parameter sets (alternator-regulator and genset-controller) are necessary

A Rolls-Royce solution

Application data [1]

Engine

Manufacturer	*mtu*
Model	20V4000G34F
Type	4-cycle
Arrangement	20V
Displacement: l	95.4
Bore: mm	170
Stroke: mm	210
Compression ratio	16.4
Rated speed: rpm	1500
Engine governor	ECU 9
Max power: kWm	2590
Air cleaner	dry

Fuel system

Maximum fuel lift: m	5
Total fuel flow: l/min	27

Fuel consumption [2]	l/hr	g/kwh
At 100% of power rating:	599.1	192
At 75% of power rating:	449.3	192
At 50% of power rating:	312	200

Liquid capacity (lubrication)

Total oil system capacity: l	390
Engine jacket water capacity: l	205
Intercooler coolant capacity: l	50

Combustion air requirements

Combustion air volume: m^3/s	2.9
Max. air intake restriction: mbar	50

Cooling/radiator system

Coolant flow rate (HT circuit): m^3/hr	80
Coolant flow rate (LT circuit): m^3/hr	32.5
Heat rejection to coolant: kW	950
Heat radiated to charge air cooling: kW	410
Heat radiated to ambient: kW	105
Fan power for electr. radiator (40°C): kW	70

Exhaust system

Exhaust gas temp. (after turbocharger): °C	565
Exhaust gas volume: m^3/s	7.7
Maximum allowable back pressure: mbar	85
Minimum allowable back pressure: mbar	30

Standard and optional features

System ratings (kW/kVA)

Generator model	Voltage	Fuel consumption optimized					
		without radiator			with mechanical radiator		
		kWel	kVA*	AMPS	kWel	kVA*	AMPS
Leroy Somer LSA53.2 M12 (Low voltage Leroy Somer standard)	380 V	2488	3110	4725	2424	3030	4604
	400 V	2488	3110	4489	2424	3030	4373
	415 V	2488	3110	4327	2424	3030	4215
Marathon 1030FDL7094 (Low voltage Marathon)	380 V	2496	3120	4740	2416	3020	4588
	400 V	2488	3110	4489	2416	3020	4359
	415 V	2488	3110	4327	2416	3020	4201
Marathon 1040FDH7102 (Medium volt. marathon)	11 kV	2496	3120	164	2416	3020	159
Leroy Somer LSA54.2 XL11 (Medium volt. Leroy Somer)	11 kV	2504	3130	164	2424	3030	159

* cos phi = 0.8

1 All data refers only to the engine and is based on ISO standard conditions (25°C and 100m above sea level).

2 Values referenced are in accordance with ISO 3046-1. Conversion calculated with fuel density of 0.83 g/ml. All fuel consumption values refer to rated engine power.